clinical failure: fixed-effect meta-analysis (risk difference)

Comparison: Intervention / Control

Methods

Risk difference estimates were pooled using a fixed-effect model. A continuity correction of 0.5 was applied to individual study estimates only; the pooled Mantel-Haenszel estimate uses the original counts. Heterogeneity was assessed with Cochran's Q and I^2. Analyses were performed in RC MetaStudio 0.5.1. R packages loaded for this analysis: metafor 5.0.1, meta 8.5-0.

Results

19 studies were included (1097 participants in Intervention, 1080 in Control). The pooled risk difference was -0.030256 (95% CI [-0.055844, -0.004668], z = -2.317520, p = .020). The confidence interval excludes 0, with lower risk in Intervention than in Control. Heterogeneity: Q = 43.705892, df = 18, p < .001. I^2 = 65.7%, which may represent substantial heterogeneity (Cochrane Handbook ranges).

Table 1. Pooled effect

Effect	Estimate	95% CI	SE	z	p
Risk difference	-0.030256	-0.055844 to -0.004668	0.013055	-2.317520	.020

Table 2. Model fit

Statistic	Value
Studies (k)	19
Participants in Intervention	1097
Participants in Control	1080
Cochran’s Q	43.705892
Heterogeneity degrees of freedom	18
Heterogeneity p	< .001
I^2	65.7%

Table 3. Study results

Study	Effect	95% CI	Weight
Gonzalez	0.000000	-0.069371 to 0.069371	2.5%
Prins	-0.058528	-0.153339 to 0.036284	5.6%
Giamarellou	-0.161290	-0.302893 to -0.019688	2.8%
Maller	0.030327	-0.035823 to 0.096478	14.5%
Sturm	-0.054054	-0.143077 to 0.034969	3.3%
Marik	-0.170189	-0.267077 to -0.073301	13.8%
Muijsken	0.078571	-0.116617 to 0.273759	3.1%
Vigano	0.012958	-0.024221 to 0.050136	6.6%
Hansen	-0.283410	-0.476977 to -0.089844	2.7%
De Vries	0.035526	-0.036915 to 0.107967	7.2%
Mauracher	-0.063773	-0.117171 to -0.010375	8.5%
Nordstrom	-0.044700	-0.218599 to 0.129199	2.6%
Rozdzinski	0.003130	-0.133496 to 0.139756	6.6%
Ter Braak	0.033816	-0.076496 to 0.144128	6.5%
Tulkens	0.000000	-0.096765 to 0.096765	1.7%
Van der Auwera	-0.064516	-0.170128 to 0.041096	2.8%
Klastersky	0.320000	0.087432 to 0.552568	2.3%
Vanhaeverbeek	0.001190	-0.093327 to 0.095708	1.8%
Hollender	-0.017841	-0.065527 to 0.029845	5.2%

Study	Intervention Events	Intervention Total	Control Events	Control Total	Weight	Risk Difference M–H, Fixed, 95% CI
Gonzalez 1993	0	27	0	27	2.5%	0.000000 [-0.069371, 0.069371]
Prins 1993	3	59	7	64	5.6%	-0.058528 [-0.153339, 0.036284]
Giamarellou 1991	0	30	5	30	2.8%	-0.161290 [-0.302893, -0.019688]
Maller 1993	19	164	13	152	14.5%	0.030327 [-0.035823, 0.096478]
Sturm 1989	0	36	2	36	3.3%	-0.054054 [-0.143077, 0.034969]
Marik 1991	26	155	49	145	13.8%	-0.170189 [-0.267077, -0.073301]
Muijsken 1988	8	32	6	35	3.1%	0.078571 [-0.116617, 0.273759]
Vigano 1992	1	74	0	70	6.6%	0.012958 [-0.024221, 0.050136]
Hansen 1988	2	28	11	31	2.7%	-0.283410 [-0.476977, -0.089844]
De Vries 1990	6	80	3	76	7.2%	0.035526 [-0.036915, 0.107967]
Mauracher 1989	0	92	6	93	8.5%	-0.063773 [-0.117171, -0.010375]
Nordstrom 1990	3	29	4	27	2.6%	-0.044700 [-0.218599, 0.129199]
Rozdzinski 1993	16	71	16	72	6.6%	0.003130 [-0.133496, 0.139756]
Ter Braak 1990	10	69	8	72	6.5%	0.033816 [-0.076496, 0.144128]
Tulkens 1988	0	19	0	19	1.7%	0.000000 [-0.096765, 0.096765]
Van der Auwera 1991	0	30	2	30	2.8%	-0.064516 [-0.170128, 0.041096]
Klastersky 1977	11	25	3	25	2.3%	0.320000 [0.087432, 0.552568]
Vanhaeverbeek 1993	0	19	0	20	1.8%	0.001190 [-0.093327, 0.095708]
Hollender 1989	0	58	1	56	5.2%	-0.017841 [-0.065527, 0.029845]
Fixed-effect model (Mantel–Haenszel)		**1097**		**1080**	**100.0%**	**-0.030256 [-0.055844, -0.004668]**
Total events:	105		136			

Heterogeneity: Q = 43.705892, df = 18 (P = 0.000122); I^2 = 65.7%

Test for overall effect: Z = -2.317520 (P = 0.020475)

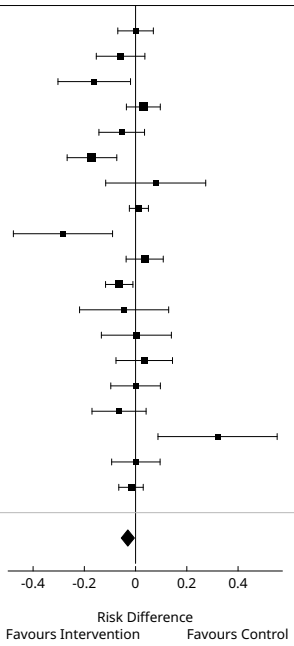

Figure 1. Forest Plot

Supplementary details

Table 4. Full-precision results

Source	Results
Summary — Model Results	Estimate: -0.030256 Lower bound (95% CI): -0.055844 Upper bound (95% CI): -0.004668 SE (outcome scale): 0.013055 p-value: 0.020475
Summary — Heterogeneity	Q: 43.705892 df: 18 p-value: 0.000122 I^2: 65.7%
Summary — Test for overall effect	Z: -2.317520 p-value: 0.020475
Weights — Study weights	Study: Gonzalez Effect: 0.000000 Lower bound: -0.069371 Upper bound: 0.069371 Weight (%): 2.5%
Weights — Study weights	Study: Prins Effect: -0.058528 Lower bound: -0.153339 Upper bound: 0.036284 Weight (%): 5.6%
Weights — Study weights	Study: Giamarellou Effect: -0.161290 Lower bound: -0.302893 Upper bound: -0.019688 Weight (%): 2.8%
Weights — Study weights	Study: Maller Effect: 0.030327 Lower bound: -0.035823 Upper bound: 0.096478 Weight (%): 14.5%
Weights — Study weights	Study: Sturm Effect: -0.054054 Lower bound: -0.143077 Upper bound: 0.034969 Weight (%): 3.3%
Weights — Study weights	Study: Marik Effect: -0.170189 Lower bound: -0.267077 Upper bound: -0.073301 Weight (%): 13.8%
Weights — Study weights	Study: Muijsken Effect: 0.078571 Lower bound: -0.116617 Upper bound: 0.273759 Weight (%): 3.1%
Weights — Study weights	Study: Vigano Effect: 0.012958 Lower bound: -0.024221 Upper bound: 0.050136 Weight (%): 6.6%
Weights — Study weights	Study: Hansen Effect: -0.283410 Lower bound: -0.476977 Upper bound: -0.089844 Weight (%): 2.7%
Weights — Study weights	Study: De Vries Effect: 0.035526 Lower bound: -0.036915 Upper bound: 0.107967 Weight (%): 7.2%
Weights — Study weights	Study: Mauracher

	Effect: -0.063773 Lower bound: -0.117171 Upper bound: -0.010375 Weight (%): 8.5%
Weights — Study weights	Study: Nordstrom Effect: -0.044700 Lower bound: -0.218599 Upper bound: 0.129199 Weight (%): 2.6%
Weights — Study weights	Study: Rozdzinski Effect: 0.003130 Lower bound: -0.133496 Upper bound: 0.139756 Weight (%): 6.6%
Weights — Study weights	Study: Ter Braak Effect: 0.033816 Lower bound: -0.076496 Upper bound: 0.144128 Weight (%): 6.5%
Weights — Study weights	Study: Tulkens Effect: 0.000000 Lower bound: -0.096765 Upper bound: 0.096765 Weight (%): 1.7%
Weights — Study weights	Study: Van der Auwera Effect: -0.064516 Lower bound: -0.170128 Upper bound: 0.041096 Weight (%): 2.8%
Weights — Study weights	Study: Klastersky Effect: 0.320000 Lower bound: 0.087432 Upper bound: 0.552568 Weight (%): 2.3%
Weights — Study weights	Study: Vanhaeverbeek Effect: 0.001190 Lower bound: -0.093327 Upper bound: 0.095708 Weight (%): 1.8%
Weights — Study weights	Study: Hollender Effect: -0.017841 Lower bound: -0.065527 Upper bound: 0.029845 Weight (%): 5.2%

Pooled Risk Difference: -0.030256 (95% CI -0.055844 to -0.004668).

References

1. Mantel-Haenszel meta-analysis: Mantel, N., & Haenszel, W. (1959). Statistical aspects of the analysis of data from retrospective studies of disease. Journal of the National Cancer Institute, 22(4), 719-748. doi:10.1093/jnci/22.4.719.

2. Common-effect sparse-data estimation: Greenland, S., & Robins, J. M. (1985). Estimation of a common effect parameter from sparse follow-up data. Biometrics, 41(1), 55-68. doi:10.2307/2530643.

3. Implementation reference: Viechtbauer, W. (2010). Conducting meta-analyses in R with the metafor package. Journal of Statistical Software, 36(3), 1-48. doi:10.18637/jss.v036.i03.